

Derivatives, stereo fixed media (2024)

Derivatives is a work that explores the disjunctions between human and machine auditory perception. It juxtaposes real-world sound objects with their computational abstractions and synthetic digital representations, guiding the listener's ear toward the boundary between the real and the simulated. This is explored through field recordings taken in Buenos Aires and Chicago, AI-synthesized speech, and machine-listening based manipulation of these materials.

This work premiered at the [2024 Festival Nueva Ópera](#) in Buenos Aires as part of a multimedia dance performance. *Derivatives* was also a finalist in the [Prix CIME 2025 Electroacoustic Music Competition](#).

Detailed Information about the work:

Enframe hollows out a selection of field recordings through source separation and psychoacoustic-based spectral decomposition algorithms, bringing forth the outer boundaries of human auditory perception in the frequency domain. This track implements these decompositions as a means of presenting a “frame” of human perception (the lowest and highest sounds we can hear) before gradually moving this “frame” inward, drawing the listener's attention to the extent of this boundary.

Converge also utilizes source separation and psychoacoustic-based spectral decomposition algorithms, but focuses primarily on the components of these field recordings that are most sensitive to the human auditory system as indicated by the Fletcher-Munson curves: 2kHz and 5kHz. This track juxtaposes the field recordings separated into various “sources” via a source separation algorithm, and the output of various analysis/resynthesis FFT algorithms that extract from the decomposed field recordings both linear and Mel Scale-quantized sinusoidal components from the decomposed field recordings; a perceptual frequency scale that approximates how humans perceive timbre. The rhythmic profile of this track is entirely dictated by onset detection applied to various field recordings used as timbral source material in the track.

Vestiges repurposes material used in *Enframe* and *Converge* as well as the raw field recordings, focusing on the human perception of amplitude. It uses a combination of transient detection and loudness-based FFT decomposition to separate transients from the source material, leaving only the loudest, sharpest moments. At moments, only the quietest moments remain, also extracted from the source material using loudness-based FFT decomposition.

Concrete mostly contains the raw field recordings themselves, revealing the source material to the listener. However, there are moments in which the spectral decompositions from *Enframe*, *Converge*, and *Vestiges* subtly blend with the raw field recordings, creating an uncanny memory of these previous materials as they are time-aligned with their source.

[ser], *[map:mapa]*, and *[systems]* all use AI-generated speech as their source material. *[ser]* repeats “ser,” the Spanish verb for “to be.” This word is generated using an AI speech model appearing as a singular voice. The model often mispronounces this word, and select pronunciations are time-stretched as a type of “zooming in.” This idea of “zooming in” is taken from my research on uncanniness in AI-generated images, as zooming in to the fine-grain details of these images is where most of the “syntheticness” of an AI-generated image is most extreme. This time-stretching results in an uncanny sound object that highlights the fact that this speech was indeed generated by synthetic means.

[map:mapa] contains two AI-generated speakers. The leftmost voice repeats a phrase from Jean Baudrillard's book *Simulacra and Simulations*, "The territory no longer precedes the map," and the right repeats the Spanish version of the phrase, "El territorio ya no precede al mapa." Many of the sounds heard here are not "real" words or syllables. I took the recordings of the English and Spanish texts, sliced them into discrete syllables, and cross-synthesized these syllables to make a “hybrid language” that contains qualities of both. This recitative begins with just the incomprehensible "hybrid" language between English and Spanish. As it progresses, the words of the phrases begin to slowly reveal themselves with each repetition. Like with *[ser]*, I use the repetition and time-stretching techniques to "zoom in" on specific words/syllables.

[systems] contains three voices generated using text from Michel Serres' *The Parasite (Volume I)*.

Panned left: *There are no systems except black boxes... boxes that function as spaces of transformation.*

Center: *Bodies are an interchange of time, traversed with signals, noises, messages and parasites.*

Panned right: *Transformation, deformation of information: A system is non-knowledge.*

By repeating select letters in the text-to-speech algorithm, the AI speech generator began to sing uncannily. Like with *[ser]* and *[map:mapa]*, I zoomed in on particular syllables and words via time-stretching.

Overall, this work reappropriates machine-listening techniques often used to convert human perception into quantitative data for machine perception within systems of control and capital. It aims to demonstrate the voids and overlaps between the human and machine perception of sound, while exploring the novel sonic potential within them.